



Cybersecurity Internship Final Summary Report

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Internship: Cybersecurity Analyst Internship

Project: React-Node-MySQL Authentication System



Objective

The objective of this internship was to assess the security of a web-based application, implement security improvements, verify their effectiveness through testing, and document the complete process.



Week 1 – Security Assessment

Tasks Performed

- Set up and explored the React-Node-MySQL application.
- Performed Cross-Site Scripting (XSS) and SQL Injection testing.
- Reviewed password storage in the database.
- Inspected network requests using Browser Developer Tools.
- Conducted an OWASP ZAP vulnerability scan.

Results

- XSS and SQL Injection attempts were unsuccessful.
- Passwords were securely stored using bcrypt hashing.
- OWASP ZAP identified missing security headers and other security improvements.

Vulnerabilities Identified

- Missing Content Security Policy (CSP)
 - Missing Anti-clickjacking Header
 - Missing X-Content-Type-Options Header
 - Server Information Disclosure (X-Powered-By)
 - HTTP used instead of HTTPS
 - JWT Authentication not implemented
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Week 2 – Security Implementation

Tasks Performed

- Implemented input validation using the Validator library.
- Verified secure password hashing using bcrypt.

- Added JWT-based authentication.
- Configured Helmet.js to improve HTTP security headers.

Results

- Invalid user inputs were successfully rejected.
- JWT tokens were generated after successful login.
- HTTP security headers were added successfully.
- Overall application security was significantly improved.

Security Fixes Implemented

- Input Validation
 - JWT Authentication
 - Helmet Security Headers
 - Secure Password Hashing
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Week 3 – Advanced Security & Verification

Tasks Performed

- Performed penetration testing to verify implemented security measures.
- Configured application logging using Winston.
- Created a security best practices checklist.
- Prepared the project for final submission.

Results

- Input validation successfully blocked invalid data.
- JWT authentication functioned correctly.
- Password hashing remained secure.
- Helmet security headers were successfully verified.
- Winston generated application logs in the `security.log` file.

Security Checklist

- ☒ Validate all user inputs
 - ☒ Hash and salt passwords
 - ☒ Implement JWT authentication
 - ☒ Secure HTTP headers with Helmet
 - ☒ Maintain application logs
 - ☒ Use HTTPS for production deployments
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Conclusion

During this three-week internship, a complete security assessment was performed on the application, followed by the implementation of important security improvements and verification through testing. The project now includes input validation, secure password hashing, JWT-based authentication, HTTP

security headers, and application logging. This internship provided valuable hands-on experience in identifying vulnerabilities, applying security best practices, and documenting cybersecurity tasks using industry-standard tools.